

Grundfos packaged lift stations

60 Hz

Installation and operating instructions



English (US) Installation and operating instructions

Original installation and operating instructions.

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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Limited warranty

Products manufactured by GRUNDFOS PUMPS CORPORATION (Grundfos) are warranted to the original user only to be free of defects in material and workmanship for a period of 24 months from date of installation, but not more than 30 months from date of manufacture. Grundfos liability under this warranty shall be limited to repairing or replacing at Grundfos option, without charge, F.O.B. Grundfos factory or authorized service station, any product of Grundfos manufacture. Grundfos will not be liable for any costs of removal, installation, transportation, or any other charges which may arise in connection with a warranty claim. Products which are sold but not manufactured by Grundfos are subject to the warranty provided by the manufacturer of said products and not by Grundfos warranty. Grundfos will not be liable for damage or wear to products caused by abnormal operating conditions, accident, abuse, misuse, unauthorized

alteration or repair, or if the product was not installed in accordance with Grundfos printed installation and operating instructions.

To obtain service under this warranty, the defective product must be returned to the distributor or dealer of Grundfos products from which it was purchased together with proof of purchase and installation date, failure date, and supporting installation data. Unless otherwise provided, the distributor or dealer will contact Grundfos or an authorized service station for instructions. Any defective product to be returned to Grundfos or a service station must be sent freight prepaid; documentation supporting the warranty claim and/or a Return Material Authorization must be included if so instructed.

GRUNDFOS WILL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES, LOSSES, OR EXPENSES ARISING FROM INSTALLATION, USE, OR ANY OTHER CAUSES. THERE ARE NO EXPRESS OR IMPLIED WARRANTIES, INCLUDING MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, WHICH EXTEND BEYOND THOSE WARRANTIES DESCRIBED OR REFERRED TO ABOVE.

Some jurisdictions do not allow the exclusion or limitation of incidental or consequential damages and some jurisdictions do not allow limit actions on how long implied warranties may last. Therefore, the above limitations or exclusions may not apply to you. This warranty gives you specific legal rights and you may also have other rights which vary from jurisdiction to jurisdiction.

2. Symbols used in this document



Warning

If these safety instructions are not observed, it may result in personal injury.



Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Note

Notes or instructions that make the job easier and ensure safe operation.

3. General description

Grundfos packaged lift stations are pre-fabricated pumping stations for collection and pumping of wastewater, such as rainwater, effluent and sewage.

The collecting tank is made of PE-HD (polyethylene) and comes with discharge pipes and valves fitted.

The pipework is made of PE or stainless steel depending on the location of the lift station and the pumped liquid, and thus the selected pump.

Unless another solution is selected, the 32" tanks, the 40" tanks and the 48" tanks have a PE-HD cover, locked with a special M10 bolt.

The 67" tanks have a fibre glass cover, with locking mechanism prepared for a padlock.

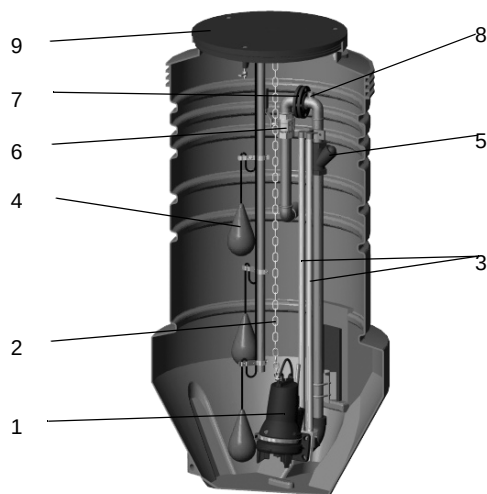


Fig. 1 Grundfos packaged lift station

Pos.	Description
1	Pump
2	Chain
3	Guide rails
4	Float switch
5	Non-return valve
6	Isolating (ball) valve
7	Flange/coupling
8	Air release
9	Cover

4. Applications

Grundfos packaged lift stations are used for collection and pumping of wastewater, such as rainwater, effluent and sewage. The pump type depends on the pumped liquid.

Wastewater is led into the collecting tank. When the liquid in the tank reaches the start level, the pump will start and pump the liquid into the sewer system.

Pumped liquids

- Drainage water
- rainwater (surface water)
- effluent
- sewage.

4.1 Liquid temperature

Maximum 104 °F (40 °C). For higher temperatures, contact Grundfos.

Caution

The liquid temperature depends on the pump selected. See the installation and operating instructions of the individual pumps.

4.2 Acids and alkalis

The lift station is resistant to strong acids and alkalis as well as solvents.

The pumps are supplied with the lift station and normally stand pH values between 4 and 10. In case of doubt, contact Grundfos.

4.3 Viscosity

Very thick wastewater must not be led into the collecting tank. See also the installation and operating instructions of the pump.

4.4 Density

Maximum 68.67 lb/ft³ (1100 kg/m³).

5. Identification

Type key

Type key, PLS.32 - PLS.48

Example	PLS	32	78	S	A	SS	SEG
Grundfos packaged lift Station							
Sump diameter [inch] *							
32:							
40:							
48:							
Length [inch]							
60:							
78:							
98:							
118:							
S:							
Simplex pump installation							
D:							
Duplex pump installation							
Installation of pump							
A:							
Auto coupling on tank bottom							
S:							
Free-standing							
Pipe material							
PE:							
Polyethylene (1.5" or 2")							
SS:							
Stainless steel, AISI 304 (1.5" or 2")							
Discharge piping is with NPSM thread							
Prepared for pump type							
SEG:							
SEG							

5.1 Nameplate

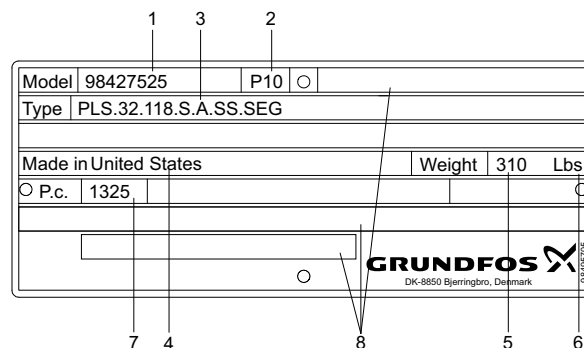


Fig. 2 Nameplate

5.2 Key to nameplate

Pos.	Description
1	Product number
2	Product site code
3	Type designation
4	Country of origin
5	Weight
6	Weight unit
7	Production code - YYWW of production
8	Not filled in

Type key, PLS.67

Example	PLS	67.	78.	S.	A.	SS.	DC.	25.
Grundfos packaged lift Station								
Sump diameter [inch] *								
67:	67							
Length [inch]								
78:								
98:								
118:								
138:								
157:								
177:								
197:								
217:								
236:								
Installation of pump								
A:	Auto coupling on tank bottom							
S:	Free-standing							
Pipe material								
PE:	Polyethylene							
SS:	Stainless steel							
Pipe design								
DC:	Direct outlet, common							
DS:	Direct outlet, separate							
GC:	Goose neck, common							
NV:	No valves							
Pipe diameter: Code number from type key / [inch]								
Stainless steel:								
25:	2.5" (63mm)							
30:	3" (75mm)							
40:	4" (110mm)							
PE:								
25:	2.5" (63mm)							
30:	3" (75mm)							
40:	4" (110mm)							
Discharge piping is with ANSI flange connection								

6. Lift station types

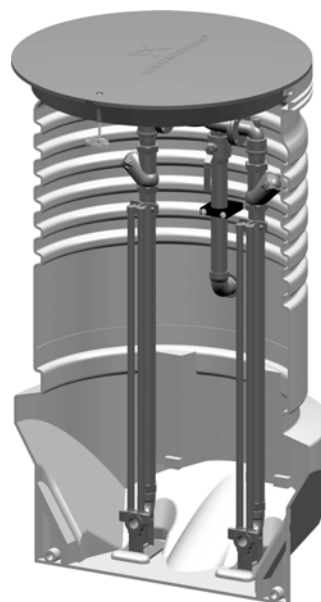
The lift stations come with pipes and valves, depending of the installation type selected. Some pumps are available with a float switch connected directly to the pump, and thus require no external or built-in controller.

The following installation drawings show four different types of lift station.



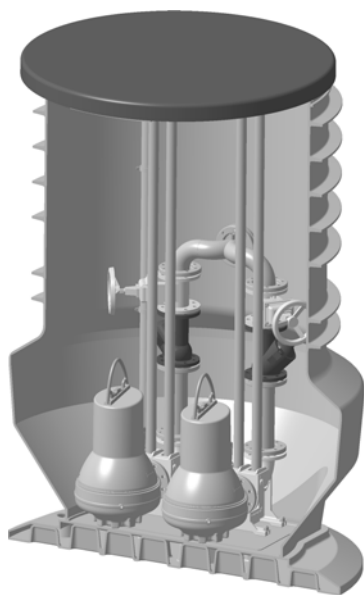
TM04 4617 1809

Lift station type PLS.32 and PLS 40 with one SEG pump on auto-coupling, goose neck outlet.



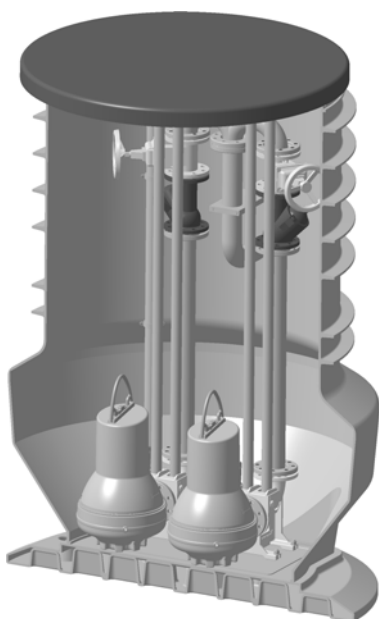
TM05 3707 1612

Lift station type PLS 48 for two SEG pumps on auto-coupling, goose neck outlet.



TM05 8451 2313

Lift station type PLS.67 with two SL solids handling pumps on auto coupling, direct outlet



TM05 8452 2313

Lift station type PLS.67 with two SL solids handling pumps on auto coupling, goose neck outlet

6.1 Cover

6.1.1 PLS32 - PLS48

Unless another solution is selected, D32 to D48 lift stations are supplied with a PE-HD cover locked with a special M10 bolt.



Fig. 3 Cover

TM02 9499 0805



TM02 9498 0805

Fig. 4 Locked cover

6.1.2 PLS.67

Covers for PLS.67 lift stations come with locking mechanism prepared for a padlock.



Warning

The lift station must be locked to prevent unwanted access.

6.2 Vent pipe

We recommend to provide the lift station with a vent pipe.



Fig. 5 lift station with vent pipe

TM02 9496 2704

6.3 Cables

The cables to the level switches and the pump can be led into the lift station in different ways. Often the cables are led into the lift station through a cable entry in the side of the lift station. If the lift station is equipped with a vent pipe, the pipe can be used as cable entry.

7. Technical data

7.1 Weight

7.1.1 PLS32 - PLS48

PLS.48.118.D.A.SS.SEG is the heaviest standard lift station, weighing 569 lb (258 kg).

7.1.2 PLS 67

PLS.67.236.D.A.SS.SL is the heaviest standard lift station, weighing approximately 518 lb (235 kg).

8. Sizing

The sump volume depends on the wastewater flow rate and the pump performance.

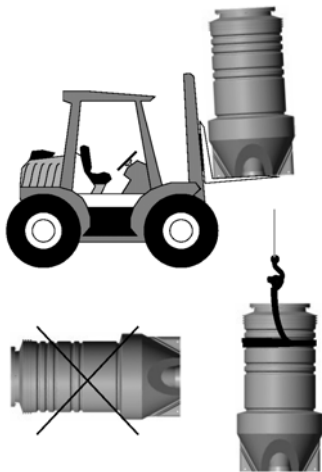
If the liquid in the lift station is static during long periods, sediments may build up in the sump. The pump should therefore be started at least twice every 24 hours.

9. Transportation

When handling and transporting the lift station at low temperatures, take into consideration that the lift station impact resistance is low.

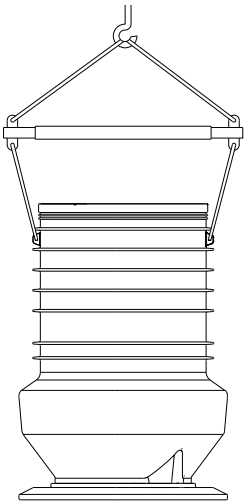
Precautions in order to avoid damage to the lift station during handling/transport:

- Do not tip the lift station from the fork lift.
- Use lifting straps of nylon or similar materials when lifting the lift station on or from a fork lift or moving it.
- Do not drag the lift station over the ground.
- Always use a suitable means of transport.
- The lift station must not come in contact with sharp edges.
- Fix the lift station securely during transport.



TM02 9340 2404

Fig. 6 Lifting and handling the lift stations (PLS32 - PLS48)



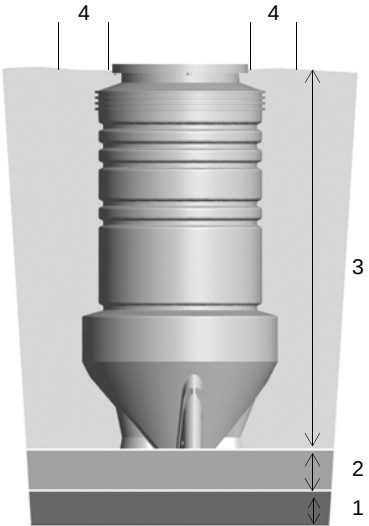
TM04 9511 4410

Fig. 7 Lifting the lift station PLS.67



Warning
Observe local regulations when lifting and handling lift stations.

10. Installation



TM02 9362 2404

Fig. 8 Schematic installation drawing

Pos.	Description
1	Foundation layer
2	Bedding layer
3	Backfill, compacted in layers of maximum 11.8 inches (300 mm)
4	Distance of 19.7 inches (500 mm) from cover where heavy traffic loads must not occur

The location of the lift station must be selected so that its establishment does not damage other constructions. Conversely, these must not damage the buried lift station.

Caution

Foundation layer

- The bottom of the excavation for a lift station must have a foundation layer if soil analyses and information about lift station load shows that the subsoil is not capable of bearing the load. The bedding layer, on which the lift station is placed, is not regarded as a foundation layer.
- The foundation layer can be made after the excavation, by laying a stable layer of suitable base gravel or similar material and compacting it in layers of maximum 11.8 inches (300 mm) (corresponding to approx. 7.9 inches (200 mm) after compaction). Such a foundation layer is also required, if the excavation has become too deep by mistake.

Bedding layer

- The material for the bedding layer must be compactible and composed so that neither its properties nor compaction will cause damage to the lift station.
- Grains larger than 0.63 inch (16 mm) must not be present.
- The content of grains between 0.31 inch (8 mm) and 0.63 inch (16 mm) must not exceed 10 %.
- The material must not be frozen.
- Sharp flint etc. must not be present.
- The bedding layer must usually have a thickness of 3.9 inches (100 mm).
If the foundation layer meets these requirements, a bedding layer is not required.

Backfill

- The backfill must provide sufficient support of the lift station on all sides and ensure that the load can be transferred without detrimental point impacts or similar impacts.
- The backfill must meet the same requirements as those for the bedding layer.
- The backfilling must be carried out so that the lift station is not damaged or deformed.
- The backfill must be compacted in layers of maximum 11.8 inches (300 mm) (corresponding to approx. 7.9 inches (200 mm) after compaction).

Caution

Before the lift station is lowered into position, various connections must be retightened, as they may have become loose during the transport.

Note

Check the lift station for external damage before lowering it into position. When the lift station has been installed, Grundfos cannot be held responsible for possible damage.

Place the lift station on the bedding layer.

Check that the lift station is vertical.

Groundwater must not rise above the bedding layer until the backfilling has been completed.

Note

Compact the backfill under the inlet and discharge pipes properly so that they are not exposed to downward loads when the backfill is compacted. See fig. 9.

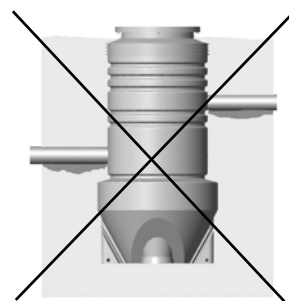


Fig. 9 Insufficient compaction under pipes

Caution

During backfilling, the cover must be on the lift station to ensure that the lift station is not deformed (becomes oval).

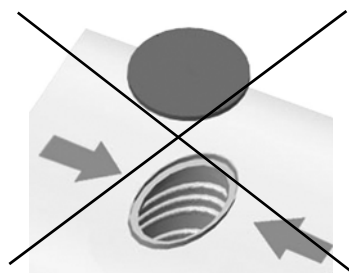


Fig. 10 Oval lift station after compaction

Caution

If the hole for the lift station is drilled, pay special attention to the compaction of the backfill to prevent uplift.

Note

Use mechanical compacting equipment and compact the backfill to 98-100 % proctor density.

Uplift prevention

- The lift station design ensures that the lift station is protected against uplift if installed correctly. Requirements for uplift prevention depend on the geotechnical conditions. They should be determined by the engineer/construction contractor and are outside Grundfos' area of responsibility.

Note

The above requirements are minimum requirements. Observe local regulations.

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TM02 9361 2404

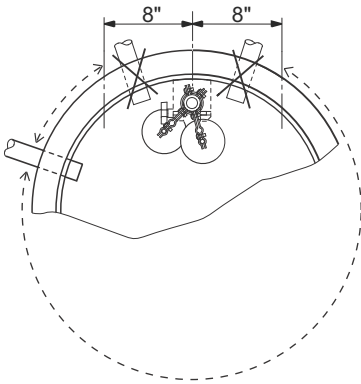
10.1 Pipe connection

Holes for inlet pipe, electrical connection and venting are to be drilled on site.
Mark the lift station where the inlet is to be. Drill the marked spot with a hole saw. Remove all burrs from the hole. The hole saw is available as an accessory. See table below.

Sleeve	Pipe diameter [inch] (mm)	Product number	Hole saw [inch] (mm)	Borehole tolerance [inch] (mm)	Product number
LM50/40	1.5 (40)	96230763	2 (51)	-0.08"/+0.04" (- 2/+ 1)	96571532
LM50/50	2 (50)	96230753	2.5 (60)	-0.08"/+0.04" (- 2/+ 1)	96571533
LM50/63	2.5 (63)	96571523	3 (75)	-0.08"/+0.04" (- 2/+ 1)	96571534
LM50/75	3 (75)	96571527	3.5 (86)	-0.08"/+0.04" (- 2/+ 1)	96571535
LM50/90	3.5 (90)	96571528	4 (102)	-0.08"/+0.04" (- 2/+ 1)	96571536
L965/110	4.3 (110)	91716040	5 (127)	-0/+0.08" (- 0/+ 2)	91713756
L965/160	6.3 (160)	91713754	7 (177)	-0/+0.08" (- 0/+ 2)	91713755
LM100/200	8 (200)	91712032	8.5 (212)	-0.08"/+0.04" (- 2/+ 1)	91712025
Pilot drill					91712026

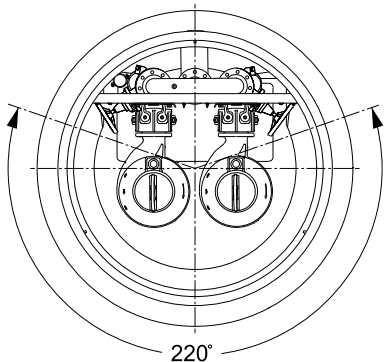
10.1.1 Locating the inlet

- When positioning the tank inlet, take the following aspects into consideration:
- The inlet must be located higher than the highest "start level" of the pump, in order to prevent backflow.
 - Avoid inflow directly over/onto pump, auto coupling, pipework, etc.
 - The tank inlet must not be located within the area shown in fig. 11, as it will disturb the function of the float switches.



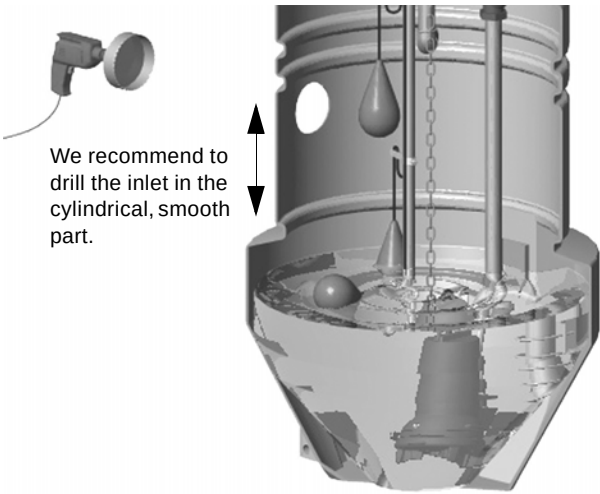
TM05 8342 2313

Fig. 11 Location of inlet, PLS.32 - PLS.48



TM05 3261 1012

Fig. 12 Location of inlet, PLS.67



TM02 9342 2404

Fig. 13 Drilling the inlet, PLS.32 - PLS.48

10.1.2 PLS.67

- Note** The hole saw must have a cup depth of minimum 3½ inches (90 mm).
- Caution** The inlet must not be drilled in the lift station close to pipes, float switches or other parts of the lift station installation.

Drilling from the outside

If the hole diameter is greater than 5.9 inches (150 mm) or the hole is to be drilled into a rib, drill in the centre of a rib. See fig 14.

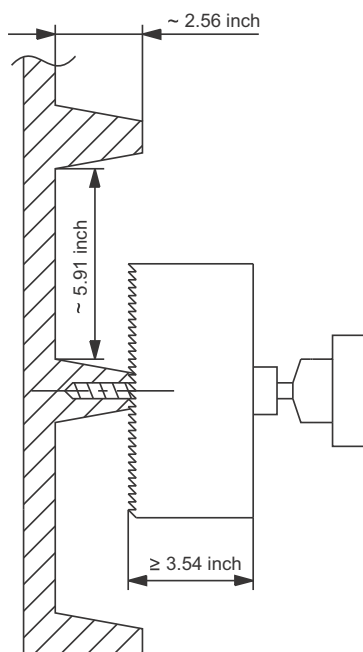


Fig. 14 Drilling from the outside

Drilling from the inside

Due to the circular lift station shape, the pilot drill must extend beyond the cutting edge of the hole saw by at least 0.9 inch (23 mm). The position of the hole is not an issue. See fig 15.

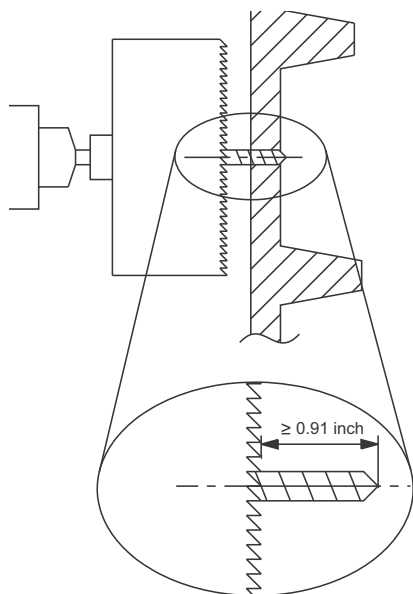


Fig. 15 Drilling from the inside

10.2 Fitting the sleeve

1. Fold the sleeve and insert it into the hole (from the outside of the lift station).
2. Unfold the sleeve so that it lies evenly in the hole.
3. **LM50/100 sleeves:** Pull the sleeve outwards until its collar touches the inside of the lift station all around the hole.
L965 sleeves: Pull the sleeve outwards until its collar touches the inside of the lift station at the positions 3 and 9 o'clock. See fig. 18.
4. Lubricate the inside of the sleeve with a water-based lubricant.

5. Chamfer the pipe and press it home.

LM50/100 sleeves: Push the inlet pipe through the sleeve. See fig. 16.

Note

As it must be possible to pull the pump up through the lift station, the inlet pipe must not project beyond the sleeve by more than 2 inches.

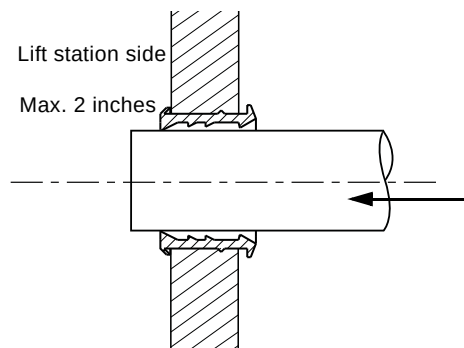


Fig. 16 Pipe with LM50/100 sleeve, seen from side

L965 sleeves: Push the inlet pipe into the sleeve and against the stop. See fig. 17.

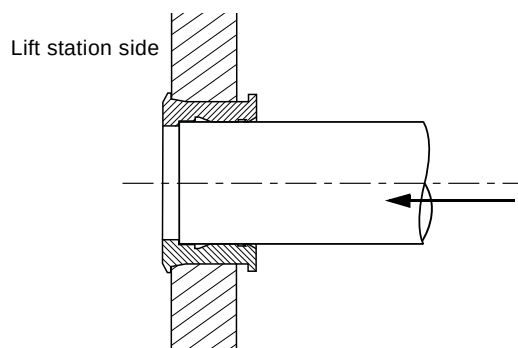


Fig. 17 Pipe with L965 sleeve, seen from side

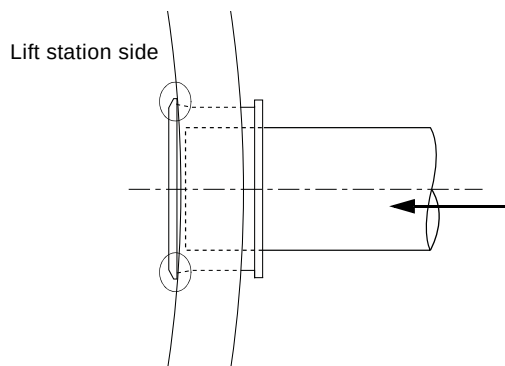


Fig. 18 Pipe with L965 sleeve, seen from top

10.3 Installation of pump

Some versions come without the pump installed. For installation and start-up of the pump, see the installation and operating instructions of the pump.

Caution

Lower the pump carefully into the lift station in order to avoid damage to pump and lift station.

TM03 3709 0806

TM03 3708 0806

TM03 3768 1006

10.4 Pipework with flange connections

If the lift station is to be installed at a temperature below 32 °F, slacken all bolts of the flange and retighten them when the lift station has been installed.

Note

This prevents stress in the pipes.

10.5 Fitting the chain

In the case of pumps on auto coupling, we recommend to fit the chain in the foremost lifting eye of the lifting bracket. The number of lifting eyes depends on the pump.



Fig. 19 Chain fitted to pump



Warning

The lifting equipment used for lifting the pump must be rated for the weight of the pump and approved according to local regulations.

11. Pump control

When installing the level switches, observe the following points:

- To prevent air intake and vibrations in the pump, fit the stop level switch in such a way that the pump is stopped before air is sucked into the pump.
- In the case of 1-pump operation, install the start level switch in such a way that the pump is started at the required level; however, the pump must always be started before the liquid level reaches the lower edge of the bottom inlet pipe.
- In the case of 2-pump operation, install the start level switches in such a way that pump 2 is started before the liquid level reaches the lower edge of the bottom inlet pipe to the lift station, and so that pump 1 is started correspondingly earlier.
- Always install the high-level alarm switch about 4 inches (100 mm) above the start level switch; however, the alarm must always be given before the liquid level reaches the inlet pipe.

For further settings, see the installation and operating instructions for the pump controller selected.

11.1 Start and stop levels

The effective volume of the lift station must be so large that the number of starts does not exceed the maximum permissible number of starts for the pump. See the installation and operating instructions of the pump.

11.2 Installation of level controller

See the installation and operating instructions of the controller.

11.3 Float switches

If float switches have been selected as level switches, they can be fitted on a tube which can be lifted out of the lift station. This ensures easy adjustment of float switches.

The bottom float switch (stop) must stop the pump before the liquid level in the lift station falls below the minimum liquid level for the pump. See the installation and operating instructions of the pump.

Note

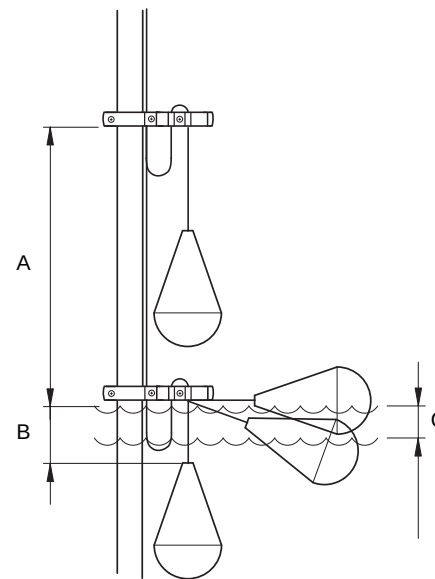


Fig. 20 Adjustment of float switches

A	Min. 11.8 inches
B	2 to 4 inches
C	Deactivation range 4.3 inches

Caution

Distance B must not be too big as otherwise the float switch may get stuck in other parts of the installation.

The lift station inlet must not be located within the area shown in fig. 21, as it will disturb the function of float switches.

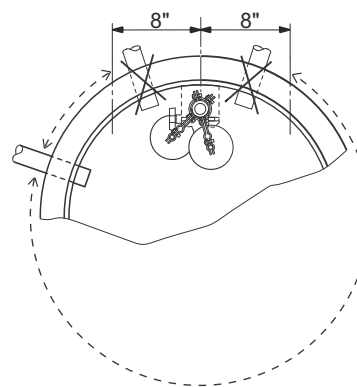


Fig. 21 Location of inlet

11.4 Pressure transducer

If a pressure transducer is used for level control, it should be installed in a riser pipe in order to prevent contamination and deposits.

11.5 Other types of level control equipment

See the installation and operating instructions of the equipment.

12. Electrical connection and start-up

See the installation and operating instructions of the pump and the controller.

Note

The controller must not be installed inside the lift station.

13. Maintenance

See the installation and operating instructions of the pump and the controller.

**Warning**

Secure the lift station cover against overload and unwanted opening and removal.

14. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.
3. If it is not possible to dispose of the product as a complete unit, the pump installation can be removed from the lift station, and the lift station can be filled and covered.

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ECM: 1125918

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